



United International University (UIU)
Department of Computer Science and Engineering
CSE 1325: Digital Logic Design, Final Exam, Fall 2025
Total Marks: 40, Duration: 2 hours

[Any examinee found adopting unfair means including copy from another examinee will be expelled from the trimester/program as per UIU disciplinary rules.]

Answer All Questions

1. A modern bank vault employs a high-security digital access mechanism to control entry. The vault controller receives a continuous serial binary data stream from a keypad encoder as authorized users enter their access credentials. The vault is programmed to unlock only when the controller detects the specific binary access code “0011” in the input data stream. When this exact sequence is detected, the system generates an unlock signal by setting the output to 1. If the correct sequence is not detected, the output remains 0, and the system continues monitoring the incoming data stream. The system must be capable of detecting the access code correctly even if overlapping sequences occur in the input stream. Do the following. (Use D flip-flops for the tasks.)

a. Design the state diagram for the sequence recognizer.

[2]

b. Create a state table for the circuit.

[2]

c. Derive the minimized logic equations for the flip-flops inputs and output.

[3]

d. Draw the circuit diagram for the sequence recognizer

[1]

2. You are designing the control unit of a digital lock used in the UIU admin office. For security reasons, the lock does not follow a normal binary counting sequence. Instead, it cycles through a secret access sequence of numbers which is $\rightarrow 3 \rightarrow 2 \rightarrow 6 \rightarrow 7 \rightarrow 0 \rightarrow 1 \rightarrow 3$.

If the system ever enters an unused or invalid state, the internal logic treats it as a don't care condition to simplify the hardware design. Do the following:

a. Prepare a state table using T Flip-Flops.

[3]

b. Minimize the functions of T inputs using Karnaugh Map.

[3]

c. Draw the circuit diagram using Flip-Flop and other basic gates.

[2]

Q(t)	Q(t+1)	T	Operation
0	0	0	No change
0	1	1	Complement
1	0	1	Complement
1	1	0	No change

3. Consider, you are voting in a national election. The Election Commission is launching an online voting system for the first time, and your team is assigned to design it.

During peak election hours, different types of events such as security alerts, login issues, and vote submissions may occur at the same time. Because some events can threaten the integrity of the election, the system must immediately prioritize the most critical event. A priority encoder is therefore used to identify the highest-priority event by producing a 3-bit output code. Whenever multiple events occur together, the system always chooses the event with the highest priority.

Priority	Event Type	Input - Output
1 (highest)	Vote Tampering Alert	T – 111
2	Server Failure	S – 101
3	Duplicate Login Detected	D – 011
4	Vote Successfully Submitted	V – 010
5 (lowest)	Voter Information Request	I – 001

Provide the followings:

- a) Derive the truth table of the priority encoder including the valid bit. [3]
 b) Derive the Boolean expressions for all the outputs. [3]
 c) Draw the logic diagram using basic gates. [2]

4. (i) In a high-security experiment room, exactly two trained persons must be present to conduct an experiment safely. At the entrance of the experiment room, there are three switches labeled A, B, and C where each person can press only one switch. A pressed switch outputs logic 1, and an unpressed switch outputs logic 0. The gate should open only when exactly two switches are pressed simultaneously. The gate must remain closed if fewer than two or all three switches are pressed. [4]

Sample input/output			
A	B	C	Status
1	1	0	Opened
0	1	0	Unopened

Design this access control system using a decoder and any other basic gate(s).

- (ii) Design a 4-to-16 decoder using 2-to-4 decoders only. You cannot use any logic gates. [4]
5. In the design of a simple digital access control system, a control signal $F(A, B, C, D)$ is used to determine when access should be granted to specific users based on a 4-bit binary input representing user ID. The access is allowed only for specific user IDs: 0, 1, 4, 5, 8, 12, and 13.

Implement the control signal function $F(A,B,C,D) = \sum m(0,1,4,5,8,12,13)$ using a 4×1 multiplexer, and any other basic gates, high/low signals as necessary. Provide

- (i) the simplified MUX inputs and [4]
 (ii) draw the circuit diagram. [4]

